

# Does energy consumption play a key role? Re-evaluating the energy consumption-economic growth nexus from GDP growth rates forecasting

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## ABSTRACT

Given the strong link between economic growth and energy consumption, we construct a new set of energy consumption indices (ECI) to forecast GDP growth rates. The findings demonstrate the strong predictive power of the new indices. In particular, the electricity consumption index from the industrial sector has an excellent predictive performance and deserves the attention of the government and academia to help formulate economic growth targets. The mixed-frequency model with the Least Absolute Shrinkage and Selection Operator approach (MIDAS-LASSO) enables the full exploitation of energy consumption information for stable forecasting performance, even in the context of crisis fluctuations and geopolitical risk shocks. Furthermore, we apply forecast encompassing tests and variable selection to statistically investigate the predictive information. By analyzing energy consumption characteristics, we contribute to the economic growth hypothesis at the level of energy consuming sectors and provide new insights into GDP forecasting.

## 1. Introduction

As an important factor in economic production, energy is widely used in all aspects of social production and consumption, and has a significant impact on global economic activities (Gong et al., 2022a). A growing energy crisis in the 1970s inspired academia and industry to study energy consumption and economic growth (see Cleveland et al., 1984; Ramanathan, 2006; Apergis and Payne, 2009; Kourtzidis et al., 2018; Raza et al., 2019; Acheampong et al., 2021; Chai et al., 2022), as energy has always been a key driver of economic development. Energy plays an important role in economic development as a vital input component for economic growth. Since Kraft and Kraft (1978) conducted pioneering research on energy consumption and economic growth, numerous studies have examined energy consumption and economic development. For example, Shahbaz et al. (2018) and Raza et al. (2019) explored the connection between energy consumption and economic development in developed countries using quantile and covariance analysis techniques, and discovered a favorable association

between the two. Wang and Wang (2020) conducted an in-depth examination into the underlying factors that impact economic growth with regards to energy consumption. More recently, Acheampong et al. (2021) found that there exists an interdependent relationship between energy consumption and economic growth.

Overall, the theoretical literature classifies the relationship between energy consumption and economic growth into four categories: growth hypothesis, conservation hypothesis, neutrality hypothesis, and feedback hypothesis. Studies in developed economies have provided empirical evidence in support of the growth hypothesis, which holds that energy consumption plays a crucial role in economic development by complementing the inputs of capital and labor in the production process. Therefore, the contribution of energy consumption to economic growth is indispensable (Cleveland et al., 1984; Apergis and Payne, 2009; Lin and Liu, 2012).<sup>1</sup> Evidence supporting the growth hypothesis is demonstrated through increased energy consumption, economic growth, and higher GDP. This suggests a positive relationship between energy consumption and economic development, providing support for

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<sup>1</sup> The remaining three hypotheses hold as follows. The conservation hypothesis states that energy conservation policies designed to reduce energy consumption and waste do not adversely affect real GDP. The neutrality hypothesis then assumes that energy consumption is a small fraction of total output and thus has little or no effect on real GDP, and that there is no causal relationship between energy consumption and real GDP. Finally, the feedback hypothesis states that energy consumption and real GDP are correlated and that there is a two-way causal relationship.

the hypothesis.<sup>2</sup> As a result, energy consumption reveals critical information that affects the variation of GDP.

Real GDP is a composite measure of an economy's output. Therefore, forecasting GDP growth is essential for determining economic policies and monitoring implementation by governments and the business community. The work is often arduous because of the difficulty of measuring GDP and the inconsistent release of data. Therefore, trustworthy GDP forecasts are urgently needed to provide policymakers and market participants with useful information to enable timely responses to economic developments. In light of existing studies examining the relationship between energy consumption and economic development, our proposal is that the variation in GDP is significantly influenced by energy consumption. How the volatility of key energy commodities, represented by oil and natural gas, propagates in the market is important for investment portfolios and will also affect the operation of the market economy (Gong et al., 2021). It must be stressed that a great deal of research has focused on examining the inherent relationship between energy consumption and economic growth, but limited research has been done in the field of economic forecasting with respect to energy consumption. Therefore, the primary goal of this study is straightforward: to analyze the relationship between energy consumption and GDP growth, with a focus on forecasting. Given that energy consumption reflects real economic activity, we aim to forecast variations in the GDP growth rate based on energy consumption.

In addition, to achieve accurate GDP forecasts, we aim to develop a novel set of energy consumption indices. The incorporation of numerous macroeconomic variables is a popular subject in the GDP forecasting literature (see Andreou et al., 2013; Marsilli, 2014; Barsoum and Stanekiewicz, 2015; Chernis and Sekkel, 2017), and it frequently results in excellent forecasting performance. Despite the improvement in prediction accuracy, the information behind the variables is rarely used to address prediction accuracy. In addition to the economic information in the variables, the superior predictive power may also be correlated with the volume of predictors. Therefore, our second motivation is to develop a collection of short and efficient predictors of the GDP growth rate based on energy consumption. And, we want to explore which energy-consuming sectors and energy sources play a key role in GDP forecasts and deserve the attention of governments and researchers. Given that GDP plays a significant role in the macroeconomic environment, we want to find out if the new indices might be a more accurate representation of real macroeconomic activity. The information behind the predictive power of the variables is again one of the main focuses of this study.

In reality, GDP is typically released on a quarterly basis, whereas predictors such as macroeconomic indicators can be accessed on a daily or monthly basis. Moreover, predictors are susceptible to shifts in the economic climate, and their performance is prone to change under various business cycle stages or shocks. In order to effectively identify the information of predictors, even in the context of complex changes in the economic environment, we want to apply a set of models that can accurately match dependent and independent variables with different frequencies. By performing this, we can accurately predict GDP growth even in the presence of complex changes in the economic environment.

In this context, our contribution is threefold. First, unlike the vast majority of existing literature, which focuses on analyzing the relationship between energy consumption and economic growth, this study takes a unique approach by examining the impact of energy consumption on economic growth from a predictive perspective. Specifically, we employ the concept of energy consumption change rate as a metric to

explore this relationship, marking the first time that this approach has been employed in this context. We concentrate on the change rate of primary and secondary energy consumption from the commercial, residential, and industrial sectors to develop a set of energy consumption indices (ECI) after eliminating seasonal impacts. This new set of forecasting indicators can comprehensively reflect macroeconomic activity, have excellent predictive value for GDP growth rate forecasts, and also provide an innovative perspective for further exploration of the energy consumption and economic growth nexus.

Second, with only nine sub-indices, our newly developed index is more condensed than popular macroeconomic variables used for GDP forecasting and can be used as an option in the future for a large number of common macroeconomic variables. Importantly, the newly established ECI has outstanding GDP growth prediction capabilities. Among the nine indices, the energy consumption index from the industrial sector (ELI) constantly has the most consistent and superior forecasting ability. Moreover, we find that energy consumption in the industrial sector plays a key role in GDP forecasting. In comparison to the single-factor mixed-data sampling (MIDAS) model, the combination forecasting models, and the dimension reduction models, the MIDAS-LASSO model stands out as the top-performing methodology. This is because the MIDAS technique can accurately match confounding variables, and the LASSO technique can efficiently identify predictive information. Our pioneering application of the MIDAS-LASSO model to explore the relationship between energy consumption and economic growth has yielded impressive results, especially in its ability to predict GDP growth rates with exceptional accuracy. This result indicates that the MIDAS-LASSO model accurately identifies the energy consumption information, and is the most reliable model for GDP growth prediction. A set of robustness tests is also presented as evidence of the effectiveness of the new findings, including the alternative lag order, alternative GDP benchmark, and alternative in-sample length.

Finally, this study is a valuable refinement and addition to the energy consumption literature supporting the economic growth hypothesis, creating a connection between economic growth studies and economic forecasting studies. By constructing a condensed collection of energy consumption indices, we are able to achieve GDP growth rate forecasts. The in-sample analysis also confirms the reliability of the growth hypothesis in the US GDP forecasting domain. Notably, we examine the predictive potential of the new indices in terms of their economic and statistical importance, and we delve into the mechanisms behind their predictive performance. Under the circumstances of a dynamic economic climate (recessions, power crisis, financial crisis, and COVID-19 pandemic), the new index has also demonstrated great forecasting ability. When compared to Fisher et al.'s (2022) newly built Macroeconomic Attention Index (MAI), ECI has superior predictive potential for GDP growth, showing that it is a more thorough reflection of the macroeconomic environment. We further investigate the predictive power of the index and the model in the context of fluctuating geopolitical risks and economic volatility, and both yield excellent results.

Our study is inspired by Kourtzidis et al. (2018), but has the following unique features. First, in contrast to Kourtzidis et al.'s (2018) use of the threshold cointegration method to explore the asymmetric dynamics between economic growth and energy consumption, our study takes a distinct approach by prioritizing the predictive capabilities of energy consumption on GDP growth rates within the context of the broader link between energy consumption and economic growth. Second, Kourtzidis et al. (2018) found that the energy consumption increase at the national level would lead to the increase in economic growth. By creating a new set of energy consumption indices, we have established that these metrics exhibit exceptional predictive performance in predicting the GDP growth rate. This reinforces the notion of the significant role played by energy consumption in driving economic development, lending support to the hypothesis of a strong positive relationship between energy consumption and economic growth. Finally, we use the machine learning method, that is, the MIDAS-LASSO model, to

<sup>2</sup> According to the recent report from the U.S. Energy Information Administration (EIA), as population and economic growth outpace efficiency, and economic growth in turn drives energy growth, U.S. energy consumption will exceed its projected growth rate by 2050 (see, <https://www.eia.gov/todayinenergy/detail.php?id=51478>).

effectively identify the prediction information from energy consumption, which is also an extension of the research of Kourtzidis et al. (2018).

There are several important findings from this study. First, from a predictive perspective, we comprehensively examine the relationship between energy consumption and economic growth. We construct a new set of energy consumption indices and find that the new indices have excellent predictive power for GDP growth rates. In particular, ELI exhibits the strongest predictive power. Second, the MIDAS-LASSO model continually outperforms alternative forecasting models and shows a considerable improvement in out-of-sample  $R^2$  (hereafter,  $R^2_{OOS}$ ) values. Even in recessions and crises, or during periods of fluctuating geopolitical risk and economic volatility, the MIDAS-LASSO model based on ECI has useful applications. Therefore, MIDAS-LASSO is a reliable model that can effectively identify energy consumption information and thus achieve accurate GDP prediction. Importantly, this work provides a novel, concise, and effective set of predictors for GDP forecasting that more accurately reflects actual macroeconomic conditions and provides new evidence for GDP forecasting. In order to accurately track real economic growth and inflation, both government officials and market participants must carefully monitor data related to energy consumption. In particular, changes in electricity consumption from the industrial sector, which plays a key role in GDP forecasts, deserve attention. It is also the first study to forecast GDP from the impact of geopolitical shocks. Our study adds to the existing body of literature and provides enhanced insight into the intricate connection between economic growth and energy consumption.

The remainder of this paper is structured as follows. Section 2 shows the methodology, providing an overview of both the ECI construction method and the prediction method. Section 3 then provides a detailed introduction to the data. Section 4 presents the main empirical findings, complemented by multiple robustness tests and a discussion. The findings of this study are further explored in Section 5. Section 6 concludes.

## 2. Methodology

### 2.1. ECI constructions

To determine whether energy consumption predicts GDP growth rates, we divide it into three main energy-consuming sectors: residential, commercial, and industrial, and evaluate it in broad categories of primary and secondary energy.<sup>3</sup> Primary energy is generally categorized into renewable and non-renewable based on whether it can be regenerated in the short term after consumption. In addition, we apply the most widely used secondary energy, electricity, as a proxy to represent real manufacturing and operating activities. Specifically, we constructed nine energy consumption indices (ECI) based on the change rate in energy consumption:

$$c_t = \frac{x_t - x_{t-1}}{x_{t-1}} \quad (1)$$

where  $c_t$  is the energy consumption change rate,  $x_t$  and  $x_{t-1}$  are the actual consumption in month  $t$  and month  $t-1$ , respectively. It is important to note that we deal with seasonal effects to avoid prediction errors.<sup>4</sup> Specifically, in order to remove the effect of this seasonality on the law of the data itself, the data is seasonally adjusted using the commonly used census  $X-12$  additive model. The additive model is formulated as follows:

$$y_t = T_t + S_t + e_t \quad (2)$$

where  $y_t$  is the actual series value,  $T_t$  is the trend term,  $S_t$  is the seasonal term, and  $e_t$  is the random term. Therefore, we seasonally treat the actual monthly energy consumption and retain the trend term.

### 2.2. Predictive models

#### 2.2.1. MIDAS models

To investigate the various monthly predictors that contribute to quarterly GDP growth rates, we expand the MIDAS technique formally suggested by Ghysels et al. (2007). Through this approach, it becomes possible to explain the variation in low-frequency variables (such as quarterly GDP growth rates,  $y_t$ ) by analyzing the influence of high-frequency variables (such as monthly energy consumption indices,  $x_t$ ). As a result, it is possible to simply correlate high and low frequency variables and reduce the number of estimated parameters. The general form of the MIDAS model is as follows:

$$y_t = \alpha_0 + \alpha_1 A(L^{1/n}; \theta) x_{t-1}^{(n)} + \varepsilon_t^{(n)} \quad (3)$$

where  $n$  is the number of observations,  $\alpha_0$  and  $\alpha_1$  denote the parameters to be estimated, and  $\varepsilon_t$  is the random error term. We employ the polynomial  $A(L^{1/n}; \theta)$  to determine the weight of the lag operator  $L^{1/n}$  in  $x_{t-1}^{(n)}$ . In accordance with Marsilli (2014) and Barsoum and Stankiewicz (2015), we estimate the MIDAS model using the two-parameter exponential Almonds lag polynomial approach to guarantee that the lag coefficient is positive and fits the criterion for identifying model parameters. Correspondingly, the weight is expressed as:

$$w(k; \theta) = \frac{e^{(\theta_1 k + \theta_2 k^2)}}{\sum_{k=1}^K e^{(\theta_1 k + \theta_2 k^2)}} \quad (4)$$

To assess the predictive power of different predictors, we construct a single-factor MIDAS model, denoted as:

$$y_t = \alpha_0 + \alpha_1 A(L^{1/n}; \theta^a) x_{a,t-1}^{(n)} + \alpha_2 A(L^{1/n}; \theta^b) x_{b,t-1}^{(n)} + \varepsilon_t^{(n)} \quad (5)$$

#### 2.2.2. Combination approach and dimension reduction models

In light of shifting market variables and model uncertainty, the out-of-sample forecasting accuracy of single-factor models is quite unstable (see Avramov, 2002; Rapach et al., 2010). Therefore, it is risky to rely solely on the predictions of a single-factor model. We will further examine and evaluate whether the combination approach may yield more precise predictions. First, we consider five popular combination methods proposed by Rapach et al. (2010), namely, mean, median, trimmed mean (TMC), DMSPE (1.0), and DMSPE (0.9). The combination forecasting value  $\widehat{y}_{C,t}$  is constructed from a single forecast  $\widehat{y}_{q,t}$ , expressed as:

$$\widehat{y}_{C,t} = \sum_{q=1}^Q w_{q,t} \widehat{y}_{q,t} \quad (6)$$

where  $w_{m,t}$  is the weight of the combination.<sup>5</sup>

The relative wealth of data information not only increases the number of viable predictors, but also the difficulty of prediction. In light of this, we apply two novel dimension reduction methods, scaled principal component analysis (SPCA) and partial least squares (PLS), to the original predictors in order to produce more accurate predictions. In comparison to the earliest principal component analysis (PCA) approach, SPCA can assess each predictor's predictive power to carry out scaling optimization and can uncover predictors that are especially helpful for predictive statistical aims (Huang et al., 2022). In comparison, the principal component information may be extracted using the

<sup>3</sup> We collect the energy consumption statistics from the U.S. Energy Information Administration (EIA).

<sup>4</sup> Hunt et al. (2003) and Wang and Wang (2020) revealed that energy demand has high seasonality, and energy consumption also has seasonal changes.

<sup>5</sup> Rapach et al. (2010) and Zhang et al. (2019) provide a comprehensive description of the five combinations. The process of constructing these combinations is well documented in their work.

PLS approach developed by Kelly and Pruitt (2013), as well as the information and noise in the variables. These two techniques achieve dimension reduction through regression as follows:

$$\xi_i P_{i,t} = \zeta_i' D_t^{\text{SPCA}} + q_{i,t} \quad (7)$$

$$P_{i,t} = \rho_t + \widehat{D}_t^{\text{PLS}} + e_{i,t} \quad (8)$$

where  $D_t^{\text{SPCA}}$  and  $D_t^{\text{PLS}}$  denote SPCA and PLS diffusion indices, respectively.

Moreover, in this study, the dimension reduction method is integrated with the MIDAS model to construct the MIDAS-SPCA and MIDAS-PLS models for different frequency variables as follows:

$$y_t = \alpha_0 + \alpha_1 A(L^{1/n}; \theta) P_{i,t}^{\text{SPCA}} + \alpha_2 A(L^{1/n}; \theta) x_{t-i} + \varepsilon_t \quad (9)$$

$$y_t = \alpha_0 + \alpha_1 A(L^{1/n}; \theta) P_{i,t}^{\text{PLS}} + \alpha_2 A(L^{1/n}; \theta) x_{t-i} + \varepsilon_t \quad (10)$$

where  $P_{i,t}^{\text{SPCA}}$  and  $P_{i,t}^{\text{PLS}}$  represent the principal components extracted from the predictors by the SPCA and PLS methods, respectively.

### 2.2.3. MIDAS model combined with LASSO

The Least Absolute Shrinkage and Selection Operator (LASSO) approach, initially presented by Tibshirani (1996), has become increasingly relevant in finance forecasting over the years (see Chincio et al., 2019; Zhang et al., 2019; Ma et al., 2022). This approach provides accurate discriminative predictive information. In this study, we construct the MIDAS-LASSO model in order to maintain crucial information from the variables while dealing with mixed frequency data. The MIDAS-LASSO model, in particular, constrains the prediction coefficients of the uninformative factors to zero while preserving the important variables, thus reducing the dimensionality of the explanatory variables. Its objective function is:

$$\min_{\alpha_0, \alpha, \theta} \sum_t \left( V A_t^Q - \alpha_0 - \sum_{i=1}^{n_M} \alpha_i^M A_i^M(L^{1/n}; \theta) x_{t,i}^M - \sum_{i=1}^{n_Q} \alpha_i^Q A_i^Q(L; \theta) x_{t,i}^Q \right)^2 + \lambda \sum_{i=1}^n |\alpha_i| \quad (11)$$

where  $\lambda$  is the regularization coefficient that controls the penalty intensity and variable selection.  $x_{t,i}^M$  denotes the  $i$ -th monthly variable,  $i = 1, \dots, n_M$ ; while  $x_{t,i}^Q$  denotes the  $i$ -th quarterly variable,  $i = 1, \dots, n_Q$ .

According to Liu et al. (2021), we adopt the time series cross-validation method to select the optimal penalty coefficient  $\lambda$ . First, we distribute all  $\lambda_j$  equally in the interval between their upper and lower bound. Second, we divide the target of total sample  $\{y_t\}_{t=1}^T$  into the training set  $\{y_t\}_{t=1+K}^{K+s}$  and the test set  $y_{K+s+1}$ , where  $s = 0, \dots, T - k - 1$ . Finally, we calculate the mean square error (hereafter, MSE) value, expressed as:

$$\text{MSE}_{\lambda_j} = \frac{1}{T-K} \sum_{s=0}^{T-k-1} (y_{K+s+1} - \widehat{y}_{K+s+1})^2 \quad (12)$$

In this case, the MSE at the minimum value corresponds to the optimal  $\lambda$ .

### 2.3. Forecast strategy

When studying GDP growth, models that accurately predict economic performance are of greater interest to policymakers and investors. This serves the purpose of effectively monitoring and controlling economic development. As a result, our analysis significantly focuses on evaluating out-of-sample prediction performance in addition to conducting rigorous in-sample evaluation. In terms of data completeness and availability, the sample spans from February 1973 to December

2021 and contains 587 monthly data points and 196 quarterly GDP data points.<sup>6</sup> We choose the first 180 months as the in-sample length and employ the rolling window to obtain the forecast values to avoid data overlap.

#### 2.3.1. Out-of-sample $R^2$

We employ  $R_{\text{OOS}}^2$  as the prediction evaluation criterion, which is introduced in Campbell and Thompson (2008)'s influential study. The  $R_{\text{OOS}}^2$  statistic measures the competing model's mean square prediction error (hereafter, MSPE) in comparison to the benchmark, expressed as:

$$R_{\text{OOS}}^2 = 1 - \frac{\text{MSPE}_A}{\text{MSPE}_B} \quad (13)$$

where  $\text{MSPE}_A$  and  $\text{MSPE}_B$  denote the MSPE of the competing model and the benchmark, respectively, expressed as:

$$\text{MSPE}_j = q^{-1} \sum_{i=1}^q (\text{GR}_{p+i} - \widehat{\text{GR}}_{j,p+i})^2, j = A, B \quad (14)$$

where  $p$  and  $q$  denote the in-sample and out-of-sample lengths, respectively,  $\text{GR}_{p+i}$  and  $\widehat{\text{GR}}_{j,p+i}$  denote the actual and forecasted GDP growth rate in month  $p + i$ . Additionally, to determine whether the model being analyzed provides a statistically significant improvement over the benchmark, we employ the adjusted MSPE statistic (hereafter, MSPE-adj.) introduced by Clark and West (2007), expressed as:

$$f_{p+i} = (\text{GR}_{p+i} - \widehat{\text{GR}}_{B,p+i})^2 - (\text{GR}_{p+i} - \widehat{\text{GR}}_{A,p+i})^2 + (\overline{\text{GR}}_{B,p+i} - \overline{\text{GR}}_{A,p+i})^2 \quad (15)$$

#### 2.3.2. Encompassing check

Recent studies have extensively used encompassing tests to evaluate the comparative information of different forecasting models (see Jiang et al., 2019; Zhang et al., 2019; Borup and Schütte, 2022). Inspired by this, we apply forecast encompassing tests to evaluate which energy indices contain more information and which predictive methods are superior. Mathematically, the encompassing test is:

$$\text{GR}_t = (1 - \delta) \widehat{\text{GR}}_{a,t} + \delta \widehat{\text{GR}}_{b,t} \quad (16)$$

where  $\widehat{\text{GR}}_{a,t}$  and  $\widehat{\text{GR}}_{b,t}$  represent the predicted values of the competing model and the model of interest, respectively. If  $\delta(1 - \delta) = 0$ , then the two models are inclusively linked. Specifically,  $\delta = 0$  means that the predictive information in model  $a$  is greater than that in model  $b$ , while  $(1 - \delta) = 0$  means that the predictive information in model  $b$  contains the predictive information in model  $a$ . Our study adopts the design presented in Harvey et al. (1998)'s work, which proposes a null hypothesis for examining whether the model being tested fails to include relevant information from competing models (i.e.,  $H_0 : \delta(1 - \delta) = 0$ ). If we reject the null hypothesis, it indicates that the model under examination incorporates predictive information from the competing models.

### 3. Data

This study, in line with prior research which tends to concentrate on the nexus between energy consumption and economic growth in advanced economies (see Apergis and Payne, 2009; Shahbaz et al., 2018; Raza et al., 2019), investigates the rate of GDP change in the United States:

<sup>6</sup> Available data on energy consumption began in January 1973. Since we employ the monthly change rate of energy consumption as the predictor, the actual statistics are from February 1973. We employ the popular first-order autoregressive (AR(1)) model as the benchmark for GDP forecasting (see Schumacher and Breitung, 2008; Marsilli, 2014; Chernis and Sekkel, 2017). The lag order of the prediction model is 3.

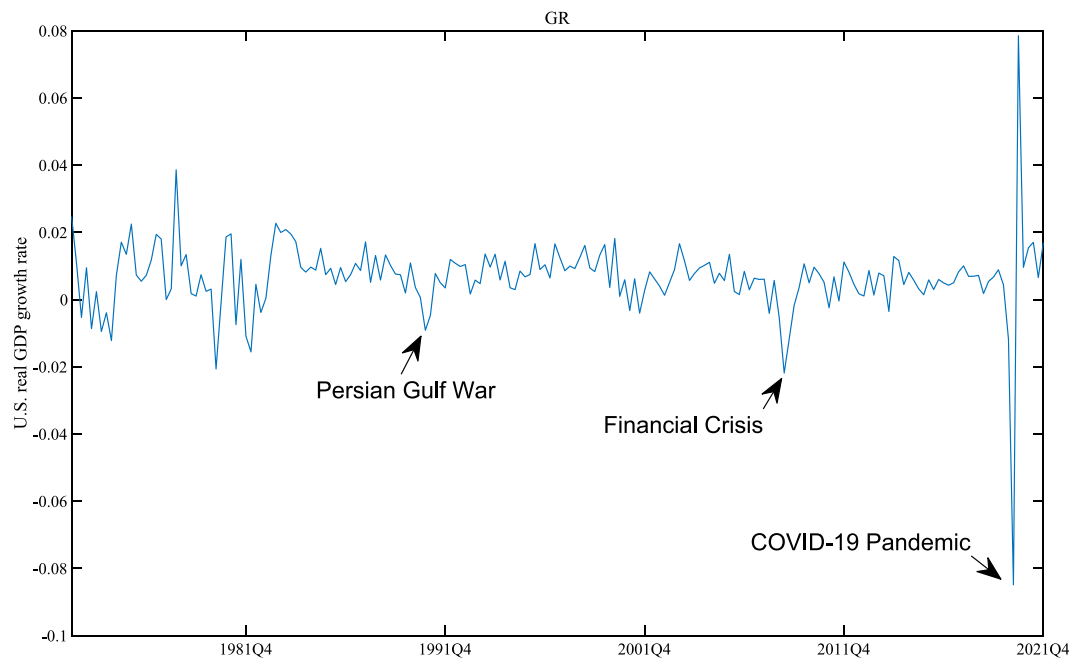


Fig. 1. GDP growth rate over the sample.

Table 1

Variable description.

| Types                                      | Variables | Detail information                                                                                  |
|--------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------|
| Forecasting target                         | GR        | Calculated as real GDP growth rates.                                                                |
|                                            | RER       | Calculated as renewable energy consumption's change rate in the residential sector.                 |
| Renewable energy                           | REC       | Calculated as renewable energy consumption's change rate in the commercial sector.                  |
|                                            | REI       | Calculated as renewable energy consumption's change rate in the industrial sector.                  |
| Primary energy                             | NOR       | Calculated as non-renewable energy consumption's change rate in the residential sector.             |
|                                            | NOC       | Calculated as non-renewable energy consumption's change rate in the commercial sector.              |
|                                            | NOI       | Calculated as non-renewable energy consumption's change rate in the industrial sector.              |
|                                            | ELR       | Calculated as the change rate of electricity sales to ultimate customers in the residential sector. |
| Secondary energy: main source, electricity | ELC       | Calculated as the change rate of electricity sales to ultimate customers in the commercial sector.  |
|                                            | ELI       | Calculated as the change rate of electricity sales to ultimate customers in the industrial sector.  |

This table presents the details of the GDP growth rate and energy consumption indices (ECI). Depending on whether the energy can exist in its natural form, the study divides it into primary and secondary energy, respectively. We further divide primary energy into renewable and non-renewable based on whether it can be regenerated in the short term. Considering the availability and representability of secondary energy statistics, we choose electricity as the representative.

$$GR_t = \frac{GDP_t - GDP_{t-1}}{GDP_{t-1}} \quad (17)$$

where  $GDP_t$  is the actual GDP value in the  $t$ -th quarter and  $GR_t$  is the corresponding GDP growth rate. Our goal is to use a new set of energy

consumption indices (ECI) to forecast and evaluate GDP growth rates. Fig. 1 depicts the progression of the GDP growth rate. What is clear is that the GDP growth rate has fluctuated significantly due to the market climate, the financial crisis and other complex reasons. It hit its nadir with the onset of the COVID-19 pandemic. In light of this, we focus on Section 5 on the predictive performance throughout the crisis.

Table 1 details the GDP growth rate and the energy consumption indices. Remarkably, we use the change rate of energy consumption as predictors, thereby removing the seasonal effect of energy consumption. Table 2 reports the descriptive statistical analysis. All series display spike patterns and pass the Augmented Dickey-Fuller test, allowing for empirical study.

## 4. Empirical results

### 4.1. Primary analysis

A logical prediction study should contain both in-sample and out-of-sample performance evaluations. In this study, we employ the ordinary least squares (OLS) method to generate in-sample regression estimates, as shown in Table 3. Our analysis of the newly constructed energy consumption index reveals that most indices contribute positively to the GDP growth rate. In particular, the electricity consumption indices from the commercial and industrial sectors (ELC and ELI) can positively influence the GDP growth rate at the 10% level of significance. This may be due to the dominant role of electricity consumption in the economy. As already established in the existing literature, electricity consumption plays a dominant role in the economy and its increase has a direct impact on the economic growth of a country.<sup>7</sup> It is worth noting that the energy consumption indices (REI, NOI and ELI) from the industrial sector can all

<sup>7</sup> Within the study of seven South American nations (Argentina, Brazil, Chile, Colombia, Ecuador, Peru, and Venezuela), Yoo and Kwak (2010) established a unidirectional causal correlation between electricity consumption and real GDP, whereby heightened electrical consumption had a direct impact on the economic progression of these particular countries. The in-sample results of our study reveal novel supporting information that establishes a positive correlation between electricity consumption within the United States and real GDP progression.

**Table 2**  
Descriptive statistical results.

| Variables | Mean  | Std.dev. | Skewness | Kurtosis | Jarque-Bera      | Q(3)       | Q(9)       | ADF        |
|-----------|-------|----------|----------|----------|------------------|------------|------------|------------|
| GR        | 0.007 | 0.011    | −1.492   | 28.351   | 19,529.620***    | 312.043*** | 320.624*** | −11.023*** |
| RER       | 0.002 | 0.030    | −2.020   | 46.622   | 52,638.500***    | 8.022**    | 12.323     | −26.817*** |
| REC       | 0.009 | 0.081    | 18.962   | 418.536  | 4,246,708.686*** | 2.688      | 5.742      | −25.648*** |
| REI       | 0.003 | 0.056    | 1.509    | 13.409   | 4538.200***      | 70.294***  | 98.890***  | −32.332*** |
| NOR       | 0.005 | 0.116    | 1.081    | 6.873    | 1245.927***      | 71.902***  | 75.215***  | −33.588*** |
| NOC       | 0.003 | 0.084    | 0.654    | 4.622    | 552.972***       | 69.421***  | 71.242***  | −33.568*** |
| NOI       | 0.000 | 0.033    | 0.410    | 2.886    | 215.175***       | 95.994***  | 113.042*** | −34.110*** |
| ELR       | 0.002 | 0.041    | 0.204    | 1.078    | 31.476***        | 47.140***  | 50.562***  | −28.845*** |
| ELC       | 0.002 | 0.018    | 0.007    | 1.177    | 32.710***        | 46.803***  | 57.239***  | −30.074*** |
| ELI       | 0.001 | 0.014    | −0.176   | 3.403    | 279.782***       | 22.582***  | 25.883***  | −27.970*** |

This table reports the descriptive statistical results. The sample period is from February 1973 to December 2022. Std.dev. is the standard error. We employ the Jarque-Bera test to evaluate whether the series are normally distributed. Q(3) and Q(9) are the Ljung-Box statistics of autocorrelation of order 3 and 9, respectively. ADF stands for the Augmented Dickey-Fuller statistic. \*\*\* and \*\* represent rejections of the null hypothesis at the 1% and 5% levels, respectively.

**Table 3**  
In-sample and out-of-sample performance.

| Predictors/Models                                                        | Slope coefficient | $R^2_{OOS}$ (%) | MSPE-adj. | $R^2_{RES}$ (%) | MSPE-adj. | $R^2_{EXP}$ (%) | MSPE-adj. |
|--------------------------------------------------------------------------|-------------------|-----------------|-----------|-----------------|-----------|-----------------|-----------|
| Panel A: Energy consumption indices.                                     |                   |                 |           |                 |           |                 |           |
| RER                                                                      | −0.014(−1.069)    | −1.312**        | 1.776     | 5.767**         | 1.774     | −108.369***     | 2.419     |
| REC                                                                      | −0.000(0.014)     | −2.961**        | 1.775     | 4.573**         | 1.770     | −116.897***     | 3.247     |
| REI                                                                      | 0.019(0.931)      | 13.957**        | 1.785     | 21.701**        | 1.793     | −103.163**      | 2.261     |
| NOR                                                                      | −0.004(−0.169)    | 1.196**         | 1.779     | 6.344**         | 1.778     | −76.657***      | 3.131     |
| NOC                                                                      | 0.020(0.976)      | 0.097**         | 1.782     | 4.932**         | 1.774     | −73.020***      | 3.433     |
| NOI                                                                      | 0.065(1.573)      | 10.578**        | 1.812     | 17.093**        | 1.810     | −87.946***      | 3.292     |
| ELR                                                                      | −0.148(−1.363)    | 0.763**         | 1.776     | 5.803**         | 1.775     | −75.459***      | 3.140     |
| ELC                                                                      | 0.372(1.777)*     | 15.457**        | 1.794     | 21.568**        | 1.794     | −76.960***      | 3.148     |
| ELI                                                                      | 0.246(1.658)*     | <b>25.671**</b> | 1.843     | <b>34.923**</b> | 1.848     | −114.243***     | 2.473     |
| Panel B: Combination models, dimension reduction models and MIDAS-LASSO. |                   |                 |           |                 |           |                 |           |
| MEAN                                                                     | —                 | 9.485**         | 1.795     | 15.218**        | 1.794     | −77.206***      | 3.083     |
| MEDIAN                                                                   | —                 | 2.435**         | 1.782     | 7.548**         | 1.780     | −74.891***      | 3.094     |
| TMC                                                                      | —                 | 7.615**         | 1.791     | 13.125**        | 1.790     | −75.704***      | 3.135     |
| DMSPE (1.0)                                                              | —                 | 10.208**        | 1.797     | 15.946**        | 1.796     | −76.563***      | 3.124     |
| DMSPE (0.9)                                                              | —                 | 10.554**        | 1.797     | 16.296**        | 1.796     | −76.295***      | 3.118     |
| MIDAS-PLS                                                                | —                 | 33.181**        | 1.832     | 35.897**        | 1.861     | −7.885          | 1.014     |
| MIDAS-SPCA                                                               | —                 | 25.315*         | 1.611     | 27.429*         | 1.631     | −6.658          | −0.088    |
| MIDAS-LASSO                                                              | —                 | <b>78.197**</b> | 1.733     | <b>83.635**</b> | 1.744     | −4.031***       | 2.881     |

This table reports in-sample and out-of-sample results. The sample period is from February 1973 to December 2022, while the first 180 months (15 years) are the in-sample period. The slope coefficient is the regression estimation of each predictor.  $R^2_{RES}$  and  $R^2_{EXP}$  represent the out-of-sample  $R^2$  during the recessions and expansions, respectively. The largest out-of-sample  $R^2$  values of each panel are bolded. \*\*\*, \*\* and \* represent rejections of null hypothesis at 1%, 5% and 10% levels, respectively.

positively affect the GDP growth rate and play a key role. That is, the faster energy consumption from the industrial sector, the faster GDP growth. This fits into the context of the United States as an industrialized nation that relies on industrial production. This also supports the growth theory. That is, an increase in energy consumption leads to an increase in real GDP, so that the economy is considered energy-dependent and the growth hypothesis is confirmed.

This study involves two primary objectives. The first objective involves investigating whether the recently developed ECI has the capacity to predict the GDP growth rate, while at the same time discerning the underlying information ascribed to such a prognostic capability. The second is how to fully and effectively use the forecast information from energy consumption to achieve accurate forecasts of GDP growth rates to provide a reference for economic regulation. Table 3 presents the energy consumption indices and the corresponding model forecast results, which yield significant revelations. In Panel A, we find that all of the energy indices have significant  $R^2_{OOS}$  values. This shows that the newly constructed ECI has excellent predictive power for the GDP growth rate. Notably, electricity consumption from the commercial and industrial sectors (ELC and ELI) have the largest  $R^2_{OOS}$  values, corresponding to their significant impacts in the in-sample analysis,

indicating that we select the best predictors in the ECI using LASSO. Among the others, ELI notably showcases the highest  $R^2_{OOS}$  value of 25.671%, solidifying its ability to furnish a substantial 5% level of prescience with regard to GDP growth. The significance of industrial electricity consumption is presumably what explains the excellent predictive power of ELI. First, end-use electricity involves retail sales to consumers and direct use, which benefits commercial, residential and industrial sectors, consuming the majority of direct use electricity.<sup>8</sup> Second, industry is a core part of the real economy. Production and life depend on industrial electricity consumption, driving the rapid growth of industrial electricity consumption. The dominant role of industrial electricity consumption affects modern living and economic performance, and a proper construction of an industrial electricity consumption index may therefore improve the performance of GDP projections. Notably, the energy consumption indices from the industrial sector (REI, NOI, and ELI) have a much better predictive performance than the other two sectors. This outcome confirms that fluctuations in energy consumption in the industrial sector can greatly affect economic

<sup>8</sup> <https://www.eia.gov/energyexplained/electricity/use-of-electricity.php>

**Table 4**  
Out-of-sample performance with an alternative lag order.

| Models                                                                   | $R^2_{OOS}(\%)$ | MSPE-adj. |
|--------------------------------------------------------------------------|-----------------|-----------|
| Panel A: Energy consumption indices.                                     |                 |           |
| MIDAS-RER                                                                | −0.545**        | 1.769     |
| MIDAS-REC                                                                | −10.722**       | 1.776     |
| MIDAS-REI                                                                | 4.323**         | 1.803     |
| MIDAS-NOR                                                                | 0.074**         | 1.776     |
| MIDAS-NOC                                                                | 0.134**         | 1.775     |
| MIDAS-NOI                                                                | 6.910**         | 1.834     |
| MIDAS-ELR                                                                | −0.395**        | 1.773     |
| MIDAS-ELC                                                                | 8.450**         | 1.803     |
| MIDAS-ELI                                                                | <b>19.682**</b> | 1.877     |
| Panel B: Combination models, dimension reduction models and MIDAS-LASSO. |                 |           |
| MEAN                                                                     | 6.120**         | 1.799     |
| MEDIAN                                                                   | 1.701**         | 1.779     |
| TMC                                                                      | 4.783**         | 1.793     |
| DMSPE (1.0)                                                              | 6.994**         | 1.802     |
| DMSPE (0.9)                                                              | 7.097**         | 1.800     |
| MIDAS-PLS                                                                | 31.156**        | 1.785     |
| MIDAS-SPCA                                                               | 25.323*         | 1.620     |
| MIDAS-LASSO                                                              | <b>74.276**</b> | 1.710     |

This table shows the out-of-sample results with the lag order of 6. The largest  $R^2_{OOS}$  values of each panel are bolded. \*\* and \* represent rejections of null hypothesis at 5% and 10% levels, respectively.

development and thus play a decisive role in GDP forecasts, which deserves the attention of the government and academics.<sup>9</sup> As indicated in Panel B, the efficacy of both the combination methods and the dimension reduction models remains stable throughout out-of-sample prediction. However, the enhancement over the single-factor MIDAS model is notably insufficient. In contrast, the MIDAS-LASSO model delivers a significant improvement in prediction accuracy. MIDAS-LASSO's superior forecasting performance is related to its capacity to compress coefficients while detecting the forecasting data of transitory coefficients (Chinco et al., 2019). In contrast, LASSO models employing the MIDAS technique are able to match monthly variables to quarterly variables with greater accuracy in parameter estimation and model fitting.

Studies have found that the business cycle can alter the process of economic growth and have a long-term influence on the economy (see Fatas, 2000; Martin and Ventura, 2012). As a result, our study delves into examining how diverse business cycle stages affect the capacity of energy indices to accurately predict GDP growth rates. We categorize the out-of-sample period into 347 expansions and 60 recessions in accordance with the official National Bureau of Economic Research (NBER) guidelines.<sup>10</sup> According to Table 3, we find that the  $R^2_{OOS}$  values of all forecasting models based on the ECI increased significantly during the recession. Significantly, ELI and MIDAS-LASSO outperform those during the recession, having the highest  $R^2_{OOS}$  values in Panels A and B, respectively. Despite this, the unsatisfactory prediction performance of the expansion may be due to the disparity in sample length, despite efforts to address this issue. These findings demonstrate that the new indices have more accurate application values for GDP during recessions, and may thus enable economic monitoring.

<sup>9</sup> For the reliability of this result, we also discuss the predictive effect of the total energy consumption from the residential, commercial, and industrial sectors (hereafter, RES, COM, and IND). Our findings confirm that energy consumption within the industrial sector is a contributing factor to GDP growth forecasts. The results are shown in Table A1.

<sup>10</sup> <https://www.nber.org/research/data/us-business-cycle-expansions-and-contractions>

**Table 5**  
Out-of-sample performance with nominal GDP growth rate.

| Models                                                                   | $R^2_{OOS}(\%)$ | MSPE-adj. |
|--------------------------------------------------------------------------|-----------------|-----------|
| Panel A: Energy consumption indices.                                     |                 |           |
| MIDAS-RER                                                                | −12.989**       | 1.702     |
| MIDAS-REC                                                                | −13.116**       | 1.699     |
| MIDAS-REI                                                                | 0.331**         | 1.705     |
| MIDAS-NOR                                                                | −10.890**       | 1.709     |
| MIDAS-NOC                                                                | −11.971**       | 1.711     |
| MIDAS-NOI                                                                | −1.602**        | 1.738     |
| MIDAS-ELR                                                                | −11.644**       | 1.703     |
| MIDAS-ELC                                                                | 5.288**         | 1.722     |
| MIDAS-ELI                                                                | <b>17.508**</b> | 1.780     |
| Panel B: Combination models, dimension reduction models and MIDAS-LASSO. |                 |           |
| MEAN                                                                     | −2.099**        | 1.723     |
| MEDIAN                                                                   | −9.833**        | 1.709     |
| TMC                                                                      | −4.415**        | 1.718     |
| DMSPE (1.0)                                                              | −1.674**        | 1.724     |
| DMSPE (0.9)                                                              | −0.903**        | 1.724     |
| MIDAS-PLS                                                                | 31.875**        | 1.868     |
| MIDAS-SPCA                                                               | 23.573*         | 1.632     |
| MIDAS-LASSO                                                              | <b>76.084**</b> | 1.720     |

This table shows the out-of-sample results with the nominal GDP. The largest  $R^2_{OOS}$  values of each panel are bolded. \*\* and \* represent rejections of null hypothesis at 5% and 10% levels, respectively.

## 4.2. Robustness checks

### 4.2.1. Alternative lag order

In the previous section, we explore the ability of the ECI to predict GDP growth rates. To ensure the reliability of the findings, we perform three robustness tests. According to Andreou et al. (2013) and Clark and Ravazzolo (2015), the lag order would have an impact on macroeconomic forecasts. As a consequence, we change the lag order to 6, and Table 4 shows the results. In general, the forecasting performance suffers after changing the lag order. Despite the weakened forecasting performance, both the MIDAS-ELI and MIDAS-LASSO models are able to effectively predict GDP growth rates with  $R^2_{OOS}$  values of 19.682% and 74.276%, respectively. However, the model based on the dimensionality reduction method also performs better in GDP prediction. Thus, energy index-based models continue to demonstrate superior predictive performance, which is consistent with our primary conclusion.

### 4.2.2. Alternative GDP benchmark

There is a considerable difference between nominal and real GDP due to fluctuations in commodity production and prices. To this end, we investigate whether the newly constructed indices are still predictive of nominal GDP. The findings, illustrated in Table 5, demonstrate that ELI outperforms all other energy indices. Price fluctuations make nominal GDP inherently more complex than real GDP, affecting the predictive effectiveness. In terms of prediction, MIDAS-LASSO outperforms other models. These results all corroborate our main finding.

### 4.2.3. Alternative in-sample length

Previous studies reveal that using a different forecasting window produces varied out-of-sample results (see Rossi and Inoue, 2012; Ma et al., 2019; Zhang et al., 2019; Liang et al., 2022; Ma et al., 2022). Therefore, the in-sample duration is crucial for the predictive effectiveness. The results of substituting the in-sample period with the initial 240 months are shown in Table 6. Similarly, in Panels A and B, both ELI and MIDAS-LASSO have the best forecasting performances, respectively. In conclusion, the newly developed ECI provides excellent predictive power for GDP growth rates and our results are valid.

## 4.3. Discussions

In the preceding discussion, we find that the newly developed ECI

**Table 6**

Out-of-sample performance with alternative in-sample lengths.

| Models                                                                   | $R^2_{OOS}(\%)$ | MSPE-adj. |
|--------------------------------------------------------------------------|-----------------|-----------|
| Panel A: Energy consumption indices.                                     |                 |           |
| MIDAS-RER                                                                | −6.083**        | 1.750     |
| MIDAS-REC                                                                | −5.850**        | 1.750     |
| MIDAS-REI                                                                | 6.955**         | 1.761     |
| MIDAS-NOR                                                                | −4.103**        | 1.756     |
| MIDAS-NOC                                                                | −5.250**        | 1.757     |
| MIDAS-NOI                                                                | 4.976**         | 1.775     |
| MIDAS-ELR                                                                | −5.183**        | 1.752     |
| MIDAS-ELC                                                                | 8.047**         | 1.759     |
| MIDAS-ELI                                                                | <b>21.013**</b> | 1.809     |
| Panel B: Combination models, dimension reduction models and MIDAS-LASSO. |                 |           |
| MEAN                                                                     | 3.274**         | 1.766     |
| MEDIAN                                                                   | −3.528**        | 1.756     |
| TMC                                                                      | 0.952**         | 1.761     |
| DMSPE (1.0)                                                              | 3.854**         | 1.767     |
| DMSPE (0.9)                                                              | 4.176**         | 1.766     |
| MIDAS-PLS                                                                | 28.614**        | 1.849     |
| MIDAS-SPCA                                                               | 19.546*         | 1.583     |
| MIDAS-LASSO                                                              | <b>71.886**</b> | 1.706     |

This table shows the out-of-sample results with alternative in-sample lengths (240 months). The largest  $R^2_{OOS}$  values of each panel are bolded. \*\* and \* represent rejections of null hypothesis at 5% and 10% levels, respectively.

has predictive power for GDP growth rates. As a result, we are interested in investigating the formal mechanisms for the successful prediction ability of these new metrics. We perform forecast encompassing tests for the ECI and the forecasting models, respectively. Tables 7 and 8 show the  $p$  values. The  $p$  values for the  $\delta$  column of the MIDAS-ELI model in Panel A of Table 7 are all  $<0.01$ , rejecting the null hypothesis with 99% confidence. The  $p$  values in this column are the most significant when compared to other single-factor models, demonstrating that MIDAS-ELI contains more predictive information. In Panel B, we find that the single-factor model provides minimal additional predictive information when compared to the combination models, dimension reduction models, and the MIDAS-LASSO model. On the other hand, Table 8 shows that the  $p$  value of the  $\delta$  column corresponding to MIDAS-LASSO is  $<0.1$  in both Panels A and B, rejecting the null hypothesis. As a result, the MIDAS-LASSO model incorporates far more information than the other models. These results provide a statistical rationale for the better predictive performance of ELI and MIDAS-LASSO.

In addition, we aim to investigate whether MIDAS-LASSO can capture valuable predictive data over the long term, and whether newly developed predictors have different predictive capabilities. Therefore, we plot Fig. 2, which depicts the predictive performance of the energy indices. The color of each square in the figure indicates the ability of the predictor to affect the GDP growth rate over the out-of-sample period. It is clear that all novel indices undoubtedly have the ability to predict the GDP growth rate, but this ability may occasionally fail.<sup>11</sup> In contrast, the MIDAS-LASSO model selectively and continuously selects predictors during prediction, leveraging valid predictive information from a sizable pool of predictors to achieve superior prediction accuracy compared to other models.

<sup>11</sup> Fig. A1 shows the selection results for different energy-consuming sectors. Due to the COVID-19 outbreak in 2020, the predictive power of the variables changed after a brief disappearance. MIDAS-LASSO is able to fully account for the predictive information of the variables and minimize the degradation of forecasting performance due to economic fluctuations.

**Table 7**  
Forecast encompassing test with energy consumption indices.

| Models                                                                   | MIDAS-RER    |          | MIDAS-REC    |          | MIDAS-REI    |          | MIDAS-NOR    |          | MIDAS-NOC    |          | MIDAS-NOI    |          | MIDAS-ELR    |          | MIDAS-ELC    |          | MIDAS-ELI    |          |
|--------------------------------------------------------------------------|--------------|----------|--------------|----------|--------------|----------|--------------|----------|--------------|----------|--------------|----------|--------------|----------|--------------|----------|--------------|----------|
|                                                                          | 1 − $\delta$ | $\delta$ | 1 − $\delta$ | $\delta$ | 1 − $\delta$ | $\delta$ | 1 − $\delta$ | $\delta$ | 1 − $\delta$ | $\delta$ | 1 − $\delta$ | $\delta$ | 1 − $\delta$ | $\delta$ | 1 − $\delta$ | $\delta$ | 1 − $\delta$ | $\delta$ |
| Panel A: Energy consumption indices.                                     |              |          |              |          |              |          |              |          |              |          |              |          |              |          |              |          |              |          |
| MIDAS-RER                                                                | −            | −        | 0.152        | 0.152    | 0.882        | 0.003    | 0.716        | 0.001    | 0.193        | 0.002    | 0.820        | 0.001    | 0.490        | 0.010    | 0.943        | 0.007    | 0.925        | 0.002    |
| MIDAS-REC                                                                | 0.152        | 0.109    | −            | −        | 0.941        | 0.005    | 0.934        | 0.078    | 0.667        | 0.113    | 0.889        | 0.004    | 0.705        | 0.096    | 0.958        | 0.008    | 0.935        | 0.002    |
| MIDAS-REI                                                                | 0.003        | 0.882    | 0.003        | 0.941    | −            | −        | 0.008        | 0.903    | 0.007        | 0.901    | 0.009        | 0.478    | 0.010        | 0.895    | 0.073        | 0.017    | 0.631        | 0.000    |
| MIDAS-NOR                                                                | 0.001        | 0.716    | 0.078        | 0.934    | 0.903        | 0.008    | −            | −        | 0.017        | 0.924    | 0.845        | 0.005    | 0.205        | 0.467    | 0.937        | 0.014    | 0.930        | 0.003    |
| MIDAS-NOC                                                                | 0.002        | 0.193    | 0.113        | 0.667    | 0.901        | 0.007    | 0.924        | 0.017    | −            | −        | 0.843        | 0.004    | 0.476        | 0.117    | 0.943        | 0.014    | 0.931        | 0.003    |
| MIDAS-NOI                                                                | 0.001        | 0.820    | 0.889        | 0.478    | 0.905        | 0.009    | 0.005        | 0.845    | 0.004        | 0.843    | −            | −        | 0.006        | 0.816    | 0.423        | 0.020    | 0.876        | 0.003    |
| MIDAS-ELR                                                                | 0.010        | 0.490    | 0.820        | 0.705    | 0.895        | 0.010    | 0.467        | 0.205    | 0.117        | 0.476    | 0.816        | 0.006    | −            | −        | 0.926        | 0.018    | 0.933        | 0.003    |
| MIDAS-ELC                                                                | 0.007        | 0.943    | 0.943        | 0.958    | 0.017        | 0.073    | 0.014        | 0.937    | 0.014        | 0.943    | 0.020        | 0.423    | 0.018        | 0.926    | −            | 0.733    | −            | −        |
| MIDAS-ELI                                                                | 0.002        | 0.925    | 0.002        | 0.935    | 0.000        | 0.631    | 0.003        | 0.930    | 0.003        | 0.931    | 0.003        | 0.876    | 0.003        | 0.933    | 0.003        | −        | −            | −        |
| Panel B: Combination models, dimension reduction models and MIDAS-LASSO. |              |          |              |          |              |          |              |          |              |          |              |          |              |          |              |          |              |          |
| MEAN                                                                     | 0.001        | 0.991    | 0.005        | 0.996    | 0.671        | 0.026    | 0.003        | 0.992    | 0.004        | 0.991    | 0.323        | 0.062    | 0.006        | 0.986    | 0.803        | 0.044    | 0.909        | 0.004    |
| MEDIAN                                                                   | 0.000        | 0.951    | 0.056        | 0.994    | 0.889        | 0.011    | 0.003        | 0.970    | 0.002        | 0.980    | 0.807        | 0.008    | 0.026        | 0.894    | 0.931        | 0.021    | 0.931        | 0.004    |
| TMC                                                                      | 0.001        | 0.989    | 0.009        | 0.995    | 0.768        | 0.017    | 0.003        | 0.993    | 0.004        | 0.991    | 0.527        | 0.021    | 0.008        | 0.983    | 0.870        | 0.032    | 0.912        | 0.004    |
| DMSPE (1.0)                                                              | 0.001        | 0.991    | 0.004        | 0.996    | 0.594        | 0.030    | 0.003        | 0.992    | 0.004        | 0.991    | 0.237        | 0.093    | 0.005        | 0.987    | 0.769        | 0.052    | 0.906        | 0.005    |
| DMSPE (0.9)                                                              | 0.001        | 0.991    | 0.004        | 0.996    | 0.561        | 0.035    | 0.003        | 0.993    | 0.004        | 0.991    | 0.198        | 0.116    | 0.005        | 0.987    | 0.750        | 0.056    | 0.903        | 0.005    |
| MIDAS-PLS                                                                | 0.016        | 0.253    | 0.013        | 0.252    | 0.018        | 0.223    | 0.018        | 0.249    | 0.017        | 0.247    | 0.020        | 0.235    | 0.018        | 0.258    | 0.017        | 0.206    | 0.044        | 0.216    |
| MIDAS-SPCA                                                               | 0.026        | 0.208    | 0.023        | 0.209    | 0.028        | 0.184    | 0.029        | 0.205    | 0.029        | 0.206    | 0.028        | 0.184    | 0.030        | 0.210    | 0.028        | 0.175    | 0.060        | 0.171    |
| MIDAS-LASSO                                                              | 0.002        | 0.622    | 0.001        | 0.594    | 0.001        | 0.532    | 0.002        | 0.589    | 0.002        | 0.586    | 0.001        | 0.535    | 0.002        | 0.610    | 0.001        | 0.490    | 0.001        | 0.529    |

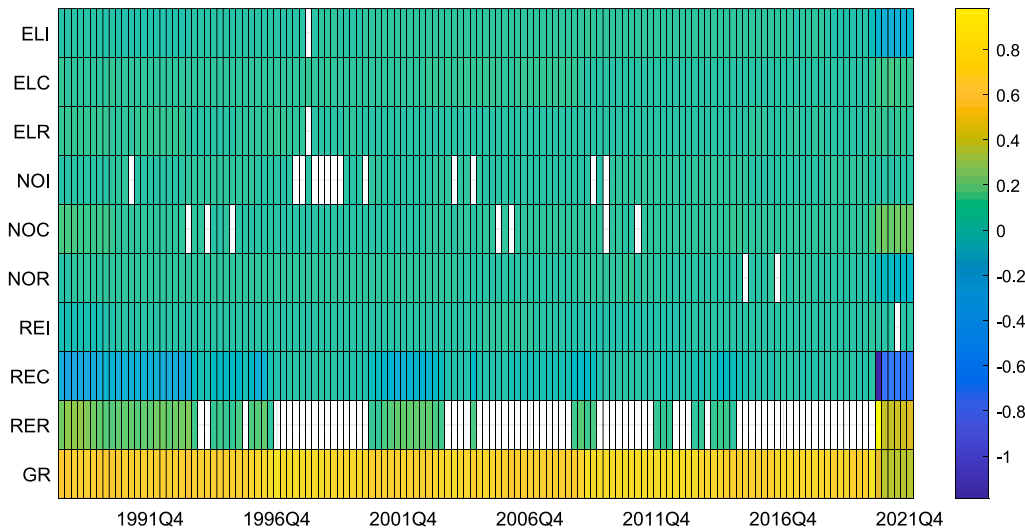
This table reports the  $p$  values of the forecast encompassing test with energy consumption indices. The  $p$  values of the “ $\delta$ ” column test the null hypothesis that the competing forecasts in the first column encompass the forecasts in the first row, while the “ $1 - \delta$ ” column tests the null hypothesis that the competing forecasts in the first row encompass the forecasts in the first column. The  $p$  values  $<0.1$  are shown in bold.

**Table 8**

Forecast encompassing test with different forecasting methods.

| Models                                                                   | MEAN         |              | MEDIAN       |              | TMC          |              | DMSPE (1.0)  |              | DMSPE (0.9)  |              | MIDAS-PLS    |              | MIDAS-SPCA   |              | MIDAS-LASSO  |              |
|--------------------------------------------------------------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|                                                                          | 1 - $\delta$ | $\delta$     | 1 - $\delta$ | $\delta$     | 1 - $\delta$ | $\delta$     | 1 - $\delta$ | $\delta$     | 1 - $\delta$ | $\delta$     | 1 - $\delta$ | $\delta$     | 1 - $\delta$ | $\delta$     | 1 - $\delta$ | $\delta$     |
| Panel A: Energy consumption indices.                                     |              |              |              |              |              |              |              |              |              |              |              |              |              |              |              |              |
| MIDAS-RER                                                                | 0.991        | <b>0.001</b> | 0.951        | <b>0.000</b> | 0.989        | <b>0.001</b> | 0.991        | <b>0.001</b> | 0.991        | <b>0.001</b> | 0.253        | <b>0.016</b> | 0.208        | <b>0.026</b> | 0.622        | <b>0.002</b> |
| MIDAS-REC                                                                | 0.996        | <b>0.005</b> | 0.994        | <b>0.056</b> | 0.995        | <b>0.009</b> | 0.996        | <b>0.004</b> | 0.996        | <b>0.004</b> | 0.252        | <b>0.013</b> | 0.209        | <b>0.023</b> | 0.594        | <b>0.001</b> |
| MIDAS-REI                                                                | 0.026        | 0.671        | 0.011        | 0.889        | <b>0.017</b> | 0.768        | <b>0.030</b> | 0.594        | <b>0.035</b> | 0.561        | 0.223        | <b>0.018</b> | 0.184        | <b>0.028</b> | 0.532        | <b>0.001</b> |
| MIDAS-NOR                                                                | 0.992        | <b>0.003</b> | 0.970        | <b>0.003</b> | 0.993        | <b>0.003</b> | 0.992        | <b>0.003</b> | 0.993        | <b>0.003</b> | 0.249        | <b>0.018</b> | 0.205        | <b>0.029</b> | 0.589        | <b>0.002</b> |
| MIDAS-NOC                                                                | 0.991        | <b>0.004</b> | 0.980        | <b>0.002</b> | 0.991        | <b>0.004</b> | 0.991        | <b>0.004</b> | 0.991        | <b>0.003</b> | 0.247        | <b>0.017</b> | 0.206        | <b>0.029</b> | 0.586        | <b>0.002</b> |
| MIDAS-NOI                                                                | <b>0.062</b> | 0.323        | <b>0.008</b> | 0.807        | <b>0.021</b> | 0.527        | <b>0.093</b> | 0.237        | 0.116        | 0.198        | 0.235        | <b>0.020</b> | 0.184        | <b>0.028</b> | 0.535        | <b>0.001</b> |
| MIDAS-ELR                                                                | 0.986        | <b>0.006</b> | 0.894        | <b>0.026</b> | 0.983        | <b>0.008</b> | 0.987        | <b>0.005</b> | 0.987        | <b>0.005</b> | 0.258        | <b>0.018</b> | 0.210        | <b>0.030</b> | 0.610        | <b>0.002</b> |
| MIDAS-ELC                                                                | <b>0.044</b> | 0.803        | <b>0.021</b> | 0.931        | <b>0.032</b> | 0.870        | <b>0.052</b> | 0.769        | <b>0.056</b> | 0.750        | 0.206        | <b>0.017</b> | 0.175        | <b>0.028</b> | 0.490        | <b>0.001</b> |
| MIDAS-ELI                                                                | <b>0.004</b> | 0.909        | <b>0.004</b> | 0.931        | <b>0.004</b> | 0.912        | <b>0.005</b> | 0.906        | <b>0.005</b> | 0.903        | 0.216        | <b>0.044</b> | 0.171        | <b>0.060</b> | 0.529        | <b>0.001</b> |
| Panel B: Combination models, dimension reduction models and MIDAS-LASSO. |              |              |              |              |              |              |              |              |              |              |              |              |              |              |              |              |
| MEAN                                                                     | –            | –            | <b>0.006</b> | 0.990        | <b>0.007</b> | 0.989        | 0.992        | <b>0.007</b> | 0.992        | <b>0.007</b> | 0.235        | <b>0.023</b> | 0.193        | <b>0.036</b> | 0.569        | <b>0.002</b> |
| MEDIAN                                                                   | 0.990        | <b>0.006</b> | –            | –            | 0.990        | <b>0.008</b> | 0.991        | <b>0.006</b> | 0.991        | <b>0.005</b> | 0.247        | <b>0.019</b> | 0.204        | <b>0.031</b> | 0.583        | <b>0.002</b> |
| TMC                                                                      | 0.989        | <b>0.007</b> | <b>0.008</b> | 0.990        | –            | –            | 0.992        | <b>0.005</b> | 0.992        | <b>0.005</b> | 0.237        | <b>0.022</b> | 0.196        | <b>0.035</b> | 0.562        | <b>0.002</b> |
| DMSPE (1.0)                                                              | <b>0.007</b> | 0.992        | <b>0.006</b> | 0.991        | <b>0.005</b> | 0.992        | –            | –            | 0.983        | <b>0.015</b> | 0.234        | <b>0.024</b> | 0.192        | <b>0.037</b> | 0.567        | <b>0.002</b> |
| DMSPE (0.9)                                                              | <b>0.007</b> | 0.992        | <b>0.005</b> | 0.991        | <b>0.005</b> | 0.992        | <b>0.015</b> | 0.983        | –            | –            | 0.234        | <b>0.025</b> | 0.192        | <b>0.038</b> | 0.566        | <b>0.002</b> |
| MIDAS-PLS                                                                | <b>0.023</b> | 0.235        | <b>0.019</b> | 0.247        | <b>0.022</b> | 0.237        | <b>0.024</b> | 0.234        | <b>0.025</b> | 0.234        | –            | –            | 0.102        | 0.686        | 0.750        | <b>0.065</b> |
| MIDAS-SPCA                                                               | <b>0.036</b> | 0.193        | <b>0.031</b> | 0.204        | <b>0.035</b> | 0.196        | <b>0.037</b> | 0.192        | <b>0.038</b> | 0.192        | 0.686        | 0.102        | –            | –            | 0.750        | <b>0.080</b> |
| MIDAS-LASSO                                                              | <b>0.002</b> | 0.569        | <b>0.002</b> | 0.583        | <b>0.002</b> | 0.562        | <b>0.002</b> | 0.567        | <b>0.002</b> | 0.566        | <b>0.065</b> | 0.750        | <b>0.080</b> | 0.750        | –            | –            |

This table reports the  $p$  values of the forecast encompassing check with the combination models, dimension reduction models and MIDAS-LASSO. The  $p$  values of the “ $\delta$ ” column test the null hypothesis that the competing forecasts in the first column encompass the forecasts in the first row, while the “1 -  $\delta$ ” column tests the null hypothesis that the competing forecasts in the first row encompass the forecasts in the first column. The  $p$  values <0.1 are shown in bold.



**Fig. 2.** Variable selection of energy consumption indices (ECI).

## 5. Further analysis

### 5.1. How does the crisis affect forecasting performance?

The leading result of this study is that the newly developed ECI, in particular the ELI index, has predictive power for the GDP growth rate. Research indicates that market volatility correlation heightens following the emergence of a significant crisis (see Wen et al., 2021). In order to assess the application of these indicators to GDP forecasting, we examine the forecasting performance during crises. Specifically, we consider the effects of the California power crisis, the financial crisis, and the COVID-

19 Pandemic.<sup>12</sup> Table 9 presents the out-of-sample results. Importantly, energy indices and forecasting models have shown further improvement in their predictive power during the crisis. In particular, the ELI and MIDAS-LASSO models have  $R^2_{OOS}$  values of 36.067% and 83.262%, respectively. As a result, the newly constructed indices have greater application value for forecasting GDP growth rates during crises, and can also play a macro-supervisory role during economic turmoil.

<sup>12</sup> Energy consumption fluctuates as a consequence of each of these crises. During the California power crisis, for example, rising electricity costs and system-wide power shortages prompted a plethora of government-imposed power restrictions, which in turn led to significant supply failures to meet demand and forced reductions in electricity consumption. The financial crisis and the COVID-19 Pandemic not only influence financial markets, but also increase economic uncertainty (Mishkin, 2009; Altig et al., 2020), which affect energy consumption.

**Table 9**

Out-of-sample performance during the crisis.

| Models                                                                   | $R^2_{OOS}(\%)$ | MSPE-adj. |
|--------------------------------------------------------------------------|-----------------|-----------|
| Panel A: Energy consumption indices.                                     |                 |           |
| MIDAS-RER                                                                | 8.476**         | 1.790     |
| MIDAS-REC                                                                | 7.952**         | 1.788     |
| MIDAS-REI                                                                | 25.170**        | 1.809     |
| MIDAS-NOR                                                                | 9.678**         | 1.797     |
| MIDAS-NOC                                                                | 8.225**         | 1.794     |
| MIDAS-NOI                                                                | 19.671**        | 1.825     |
| MIDAS-ELR                                                                | 9.178**         | 1.793     |
| MIDAS-ELC                                                                | 25.028**        | 1.813     |
| MIDAS-ELI                                                                | <b>36.067**</b> | 1.861     |
| Panel B: Combination models, dimension reduction models and MIDAS-LASSO. |                 |           |
| MEAN                                                                     | 18.211**        | 1.812     |
| MEDIAN                                                                   | 10.803**        | 1.798     |
| TMC                                                                      | 16.211**        | 1.807     |
| DMSPE (1.0)                                                              | 18.924**        | 1.813     |
| DMSPE (0.9)                                                              | 19.258**        | 1.813     |
| MIDAS-PLS                                                                | 35.616**        | 1.850     |
| MIDAS-SPCA                                                               | 27.212*         | 1.624     |
| MIDAS-LASSO                                                              | <b>83.262**</b> | 1.736     |

This table reports the out-of-sample performance during the California power crisis, financial crisis, and COVID-19 pandemic. The largest  $R^2_{OOS}$  values of each panel are bolded. \*\* and \* represent rejections of the null hypothesis at the 5% and 10% levels, respectively.

### 5.2. Are the new indicators more reflective of the macroeconomic environment?

Our initial motivation is to develop a set of energy consumption indices to predict GDP growth rates, believing that energy consumption represents the macroeconomic environment. Recently, Fisher et al. (2022) developed a new set of macroeconomic attention indices (MAI) based on investor concern theory and news. Compared to traditional macroeconomic variables, MAI can fully account for various macroeconomic risks and more accurately represent economic conditions. Previous research indicates that MAI are useful in predicting stock market returns because they reflect investors' endogenous concerns about macro news (see Ma et al., 2022). Inspired by this, we construct single-factor MIDAS models based on ECI and MAI to assess how well each reflects the macroeconomic environment and investigate its ability to predict GDP.<sup>13</sup> Table 10 summarizes the findings. In Panel A, we find that the MIDAS-monetary\_ni model (the MIDAS model built based on the monetary attention index) has the largest  $R^2_{OOS}$  value among the single-factor MIDAS models for MAI, but is inferior to MIDAS-ELI. On the other hand, Panel B demonstrates that forecasting models built on ECI outperform those built on MAI. Similarly, we plot the screening results of the LASSO technique for the predictor MAI in Fig. 3. Comparing Figs. 2 and 3, we find that the ECI contains more information about the GDP growth rate forecasts than the MAI. This suggests that energy consumption more directly reflects the current state of the economic environment and has a better predictive effect on the GDP growth rate than investors' macroeconomic concerns.

### 5.3. Geopolitical risk shocks

Recently, Caldara and Iacoviello (2022) constructed a new set of geopolitical risk (GPR) indices to reflect economic uncertainty by

<sup>13</sup> MAI data was taken from June 1980. It should be noted that we choose the same sample (1980.06–2020.12) and the same in-sample length (180 months) for the ECI and the MAI to avoid misjudgment owing to differing prediction results induced by sample length discrepancies. The MAI statistics are accessible through Jinfei Sheng's website (<https://sites.google.com/site/shengjinfei/home>), which is detailed in Fisher et al. (2022).

**Table 10**

Comparison of the out-of-sample performance of the energy consumption indices and macroeconomic attention indices.

| Energy consumption indices (ECI)                                        |                 |           | Macroeconomic attention indices (MAI) |                 |           |
|-------------------------------------------------------------------------|-----------------|-----------|---------------------------------------|-----------------|-----------|
| Predictors/Models                                                       | $R^2_{OOS}(\%)$ | MSPE-adj. | Predictors/Models                     | $R^2_{OOS}(\%)$ | MSPE-adj. |
| Panel A: Energy consumption indices and macroeconomic attention indices |                 |           |                                       |                 |           |
| MIDAS-RER                                                               | 4.124**         | 1.787     | MIDAS-credit_rating_ni                | 4.612**         | 1.794     |
| MIDAS-REC                                                               | 4.176**         | 1.786     | MIDAS-gdp_ni                          | 3.903**         | 1.792     |
| MIDAS-REI                                                               | 21.632**        | 1.799     | MIDAS-house_mkt_ni                    | 6.463**         | 1.802     |
| MIDAS-NOR                                                               | 5.740**         | 1.792     | MIDAS-inflation_ni                    | 3.204**         | 1.779     |
| MIDAS-NOC                                                               | 4.871**         | 1.795     | MIDAS-monetary_ni                     | <b>8.799**</b>  | 1.813     |
| MIDAS-NOI                                                               | 17.429**        | 1.813     | MIDAS-oil_ni                          | 7.154**         | 1.809     |
| MIDAS-ELR                                                               | 5.350**         | 1.792     | MIDAS-unemp_ni                        | 4.179**         | 1.793     |
| MIDAS-ELC                                                               | 21.065**        | 1.799     | MIDAS-usd_ni                          | 5.863**         | 1.795     |
| MIDAS-ELI                                                               | <b>31.265**</b> | 1.842     | –                                     | –               | –         |
| Panel B: Combination models, dimension reduction models and MIDAS-LASSO |                 |           |                                       |                 |           |
| MEAN-ECI                                                                | 13.975**        | 1.804     | MEAN-MAI                              | 6.902**         | 1.798     |
| MEDIAN-ECI                                                              | 7.042**         | 1.795     | MEDIAN-MAI                            | 6.108**         | 1.793     |
| TMC-ECI                                                                 | 12.051**        | 1.801     | TMC-MAI                               | 6.874**         | 1.796     |
| DMSPE (1.0)-ECI                                                         | 14.692**        | 1.805     | DMSPE (1.0)-MAI                       | 6.942**         | 1.798     |
| DMSPE (0.9)-ECI                                                         | 15.006**        | 1.805     | DMSPE (0.9)-MAI                       | 7.450**         | 1.800     |
| MIDAS-PLS-ECI                                                           | 34.917**        | 1.804     | MIDAS-PLS-MAI                         | 24.930*         | 1.609     |
| MIDAS-SPCA-ECI                                                          | 25.931*         | 1.598     | MIDAS-SPCA-MAI                        | 24.456*         | 1.584     |
| MIDAS-LASSO-ECI                                                         | <b>80.832**</b> | 1.711     | MIDAS-LASSO-MAI                       | <b>64.029**</b> | 1.699     |

This table reports the out-of-sample performance of the energy consumption indices (ECI) and macroeconomic attention indices (MAI). In order to avoid forecasting differences due to sample length variations, we take the available sample length (1980.6–2020.12) of MAI statistics as the standard, and select the first 180 months (15 years) as the in-sample length. The largest  $R^2_{OOS}$  values of each panel are bolded. \*\* and \* represent rejections of the null hypothesis at the 5% and 10% levels, respectively.

measuring geopolitical uncertainty. They found that a higher GPR index leads to a higher downside risk to GDP growth, resulting in lower GDP growth. In addition, adverse geopolitical events and threats can affect the macroeconomic environment in many ways, such as by destroying capital stock. Geopolitical risks will also affect the supply and demand of energy consumption, thus affecting energy security (Gong et al., 2022b). Inspired by this, we divide the geopolitical risk index in the out-of-sample period into strong and weak shock periods to explore the impact of geopolitical risk:

$$\begin{cases} \text{Strong shocks : if } GPR_t > \overline{GPR} \\ \text{Weak shocks : if } GPR_t < \overline{GPR} \end{cases} \quad (18)$$

where  $GPR_t$  represents the  $t$ -th GPR index, and  $\overline{GPR}$  represents the average GPR during the out-of-sample period. Table 11 reports the results. Overall, the predictive power of predictors and models is much better during periods of weak geopolitical risk shocks than during periods of strong shocks. The presence of heightened geopolitical risk can result in reduced investment and employment. This is often associated with an increased likelihood of disasters and a greater potential for downside risk, ultimately increasing the overall risk associated with economic activity and crisis events. These factors can lower the accuracy of GDP forecasts. However, during weak shocks, the newly constructed ECI has greater applications to GDP forecasting. As an extension of Caldara and Iacoviello's (2022) research, this study explores the effects

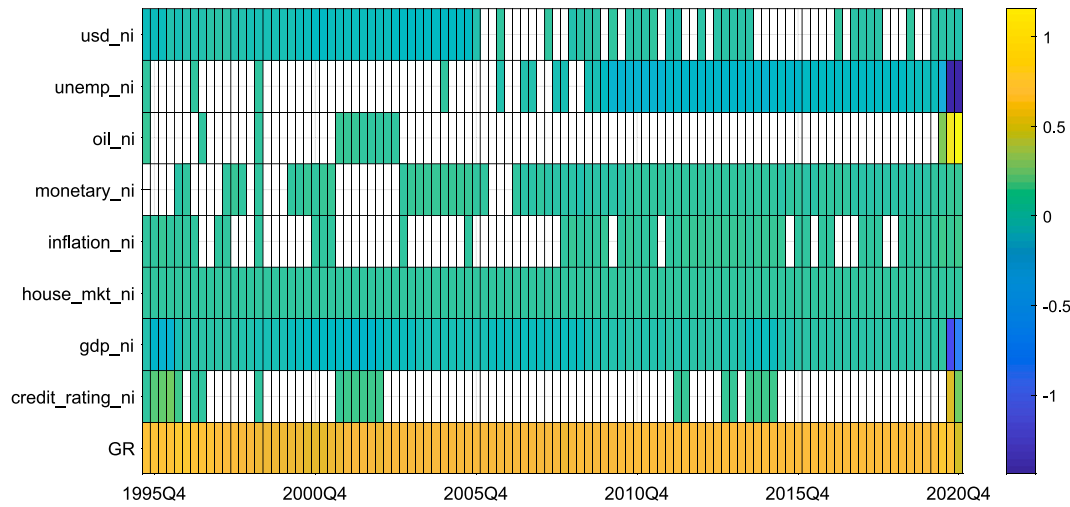


Fig. 3. Variable selection of macroeconomic attention indices (MAI).

Table 11

Out-of-sample performance during the geopolitical risk shocks.

| Models                                                                   | Strong shocks   |           | Weak shocks     |           |
|--------------------------------------------------------------------------|-----------------|-----------|-----------------|-----------|
|                                                                          | $R^2_{OOS}(\%)$ | MSPE-adj. | $R^2_{OOS}(\%)$ | MSPE-adj. |
| Panel A: Energy consumption indices                                      |                 |           |                 |           |
| MIDAS-RER                                                                | -232.252        | -2.617    | 3.682**         | 1.794     |
| MIDAS-REC                                                                | -227.155        | -2.325    | 1.888**         | 1.791     |
| MIDAS-REI                                                                | -256.082        | -2.339    | 19.797**        | 1.801     |
| MIDAS-NOR                                                                | -222.479        | -2.276    | 6.034**         | 1.795     |
| MIDAS-NOC                                                                | -229.207        | -2.308    | 5.056**         | 1.797     |
| MIDAS-NOI                                                                | <b>-203.929</b> | -1.302    | 15.217**        | 1.822     |
| MIDAS-ELR                                                                | -228.120        | -2.104    | 5.713**         | 1.791     |
| MIDAS-ELC                                                                | -244.648        | -2.304    | 21.083**        | 1.810     |
| MIDAS-ELI                                                                | -208.279        | -1.466    | <b>30.731**</b> | 1.856     |
| Panel B: Combination models, dimension reduction models and MIDAS-LASSO. |                 |           |                 |           |
| MEAN                                                                     | -217.903        | -2.197    | 14.403**        | 1.810     |
| MEDIAN                                                                   | -222.707        | -2.275    | 7.304**         | 1.797     |
| TMC                                                                      | -218.097        | -2.189    | 12.497**        | 1.806     |
| DMSPE (1.0)                                                              | -216.018        | -2.152    | 15.101**        | 1.812     |
| DMSPE (0.9)                                                              | -215.176        | -2.154    | 15.436**        | 1.811     |
| MIDAS-PLS                                                                | 0.069**         | 2.124     | 33.898**        | 1.830     |
| MIDAS-SPCA                                                               | -16.691         | -0.882    | 26.224*         | 1.616     |
| MIDAS-LASSO                                                              | <b>1.645**</b>  | 2.193     | <b>79.853**</b> | 1.730     |

This table reports the out-of-sample performance during the strong and weak geopolitical risk (GPR) shocks. The largest  $R^2_{OOS}$  values of each panel are bolded. \*\*, and \* represent rejections of the null hypothesis at the 5% and 10% levels, respectively.

of geopolitical risks on GDP forecasting and presents novel insights. It is also the first study to examine this relationship, shedding light on a previously unexplored area of research.

#### 5.4. Economic upturn and downturn

Concerns about a global economic downturn have continued to grow since the 2008 financial crisis. Commonly, the variation in several indices of economic development, such as the GDP indicator, serves to determine whether the economy is in an uptrend or downtrend (Antolin-Díaz et al., 2017). In light of this, we classify economic performance into uptrend and downtrend periods based on the positive and negative values of the GDP growth rate, denoted as:

$$\begin{cases} \text{Economic uptrend : if } GR_t \geq 0 \\ \text{Economic downturn : if } GR_t < 0 \end{cases} \quad (19)$$

Table 12

Out-of-sample performance during the economic upturn and downturn.

| Models                                                                  | Economic upturn |           | Economic downturn |           |
|-------------------------------------------------------------------------|-----------------|-----------|-------------------|-----------|
|                                                                         | $R^2_{OOS}(\%)$ | MSPE-adj. | $R^2_{OOS}(\%)$   | MSPE-adj. |
| Panel A: Energy consumption indices                                     |                 |           |                   |           |
| MIDAS-RER                                                               | 48.032**        | 1.856     | -302.703          | -1.976    |
| MIDAS-REC                                                               | 46.950**        | 1.857     | -307.817          | -1.912    |
| MIDAS-REI                                                               | 56.935**        | 1.842     | -248.556          | -2.838    |
| MIDAS-NOR                                                               | 50.929**        | 1.860     | -302.569          | -2.028    |
| MIDAS-NOC                                                               | 50.298**        | 1.862     | -306.529          | -1.987    |
| MIDAS-NOI                                                               | 56.091**        | 1.882     | -267.412          | -2.184    |
| MIDAS-ELR                                                               | 49.795**        | 1.857     | -298.723          | -2.106    |
| MIDAS-ELC                                                               | <b>62.034**</b> | 1.865     | -269.029          | -2.176    |
| MIDAS-ELI                                                               | 61.680**        | 1.890     | <b>-194.270</b>   | -2.838    |
| Panel B: Combination models, dimension reduction models and MIDAS-LASSO |                 |           |                   |           |
| MEAN                                                                    | 56.095**        | 1.867     | -275.205          | -2.158    |
| MEDIAN                                                                  | 51.354**        | 1.860     | -296.366          | -2.062    |
| TMC                                                                     | 55.234**        | 1.865     | -283.237          | -2.131    |
| DMSPE (1.0)                                                             | 56.723**        | 1.869     | -273.903          | -2.162    |
| DMSPE (0.9)                                                             | 56.969**        | 1.869     | -272.952          | -2.134    |
| MIDAS-PLS                                                               | 38.419**        | 1.808     | 1.190*            | 1.499     |
| MIDAS-SPCA                                                              | 29.232*         | 1.606     | 1.388*            | 1.451     |
| MIDAS-LASSO                                                             | <b>84.115*</b>  | 1.638     | <b>42.051</b>     | 1.042     |

This table reports the out-of-sample performance during the economic upturn and downturn. \*\*, and \* represent rejections of the null hypothesis at the 5% and 10% levels, respectively. The largest  $R^2_{OOS}$  values of each panel are bolded.

where  $GR_t$  represents the change rate of the  $t$ -th GDP. Table 12 reports the results. Overall, predictive models outperform during economic uptrends. Especially in the economic uptrend, all the predictors and models have positive and significant  $R^2_{OOS}$  values, which are far better than the general forecasting performance shown in Table 3. This result implies that the newly constructed energy consumption index has superior applications in GDP forecasting during economic uptrends. Therefore, it can be argued that energy consumption, as a complement to labor and capital, plays an important role in economic growth, directly or indirectly, in the production process, thus supporting the economic growth hypothesis. In the downturn, there are some different findings. Affected by influence of the heavy economic environment such as decreasing consumption power, except the dimension reduction models and the MIDAS-LASSO model, other models exhibit negative  $R^2_{OOS}$  values during the downturn. It is worth noting that MIDAS-LASSO

is no longer significant in the downtrend. However, the dimension reduction models have positive and significant  $R^2_{\text{OOS}}$  values, which is worth noting in GDP forecasting during the economic upturn.

## 6. Conclusions

In this study, we construct a new set of indices to explore the predictive power of energy consumption on GDP growth rates. We develop several models to establish the relationship between energy consumption and GDP growth rate, taking into account the mixed-frequency characteristics of the predictors and targets. These include single-factor and combination models, dimension reduction models, and MIDAS-LASSO models using the MIDAS technique. The results reveal that these new indices, which are strongly correlated with economic activity, have excellent predictive power on the GDP growth rate, especially the ELI index. In addition, there is a crucial aspect to consider in GDP forecasts—the energy consumption in the industrial sector. This predictor plays a crucial role and requires specific attention. In turn, the MIDAS-LASSO model stands out as the most effective predictive model that leverages the predictive information of the energy consumption index to its full potential. This agreement outperforms other predictive models in terms of performance. Several robustness tests support the reliability of this new finding. These findings support the economic growth hypothesis, showing that the expansion of energy consumption positively affects economic growth. Moreover, the usefulness of the energy consumption index in predicting GDP growth is evident, further supporting this hypothesis.

Furthermore, we investigate the information behind the energy consumption indices from economic and statistical perspectives, respectively. The forecast encompassing tests coupled with the LASSO selection plots facilitates a comprehensive and systematic analysis of the predictive effect. These analytical tools enable a more thorough evaluation and understanding of the predictions. We also examine the ability

of these new indices to predict crises, geopolitical risk shocks, economic upturns and downturns, concluding that they are more useful during crises, weak geopolitical risk shocks and economic upturns. Moreover, we compare the ECI with the MAI. Our analysis reveals that the newly developed indices exhibit superior out-of-sample performance in GDP forecasting, thus providing a more realistic representation of the macroeconomic environment. Our innovation delivers fresh insights into GDP forecasting from an energy consumption perspective, providing a new set of concise and reliable predictors for GDP forecasting. We also shed new light on economic growth studies and economic monitoring. We introduce a novel strategy for GDP forecasting that addresses the shortcomings of previous approaches that rely on macroeconomic variables. Our results support that the United States should focus on energy consumption from the industrial sector, especially electricity consumption, to monitor GDP growth and economic performance. In addition, policymakers must take into account the interplay between energy consumption and GDP growth. As energy consumption is a well-known indicator of economic performance, further investigation of its relationship to the macroeconomic environment or its application to financial market research is essential for policymakers to fully understand its implications.

### Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Appendix

Table A1 Out-of-sample performance in different energy consuming sector.

| Models                                                                   | $R^2_{\text{OOS}}$ (%) | MSPE-adj. |
|--------------------------------------------------------------------------|------------------------|-----------|
| Panel A: Energy consuming sectors.                                       |                        |           |
| MIDAS-RES                                                                | 1.665**                | 1.783     |
| MIDAS-COM                                                                | 1.537**                | 1.775     |
| MIDAS-IND                                                                | <b>19.274**</b>        | 1.824     |
| Panel B: Combination models, dimension reduction models and MIDAS-LASSO. |                        |           |
| MEAN                                                                     | 9.585**                | 1.797     |
| MEDIAN                                                                   | 3.512**                | 1.785     |
| TMC                                                                      | 3.512**                | 1.785     |
| DMSPE (1.0)                                                              | 10.140**               | 1.799     |
| DMSPE (0.9)                                                              | 10.539**               | 1.800     |
| MIDAS-PLS                                                                | 29.405**               | 1.751     |
| MIDAS-SPCA                                                               | 23.226*                | 1.565     |
| MIDAS-LASSO                                                              | <b>71.574**</b>        | 1.720     |

This table reports the out-of-sample performance from different energy consuming sector. The largest  $R^2_{\text{OOS}}$  values of each panel are bolded. \*\* and \* represent rejections of the null hypothesis at the 5% and 10% levels, respectively.

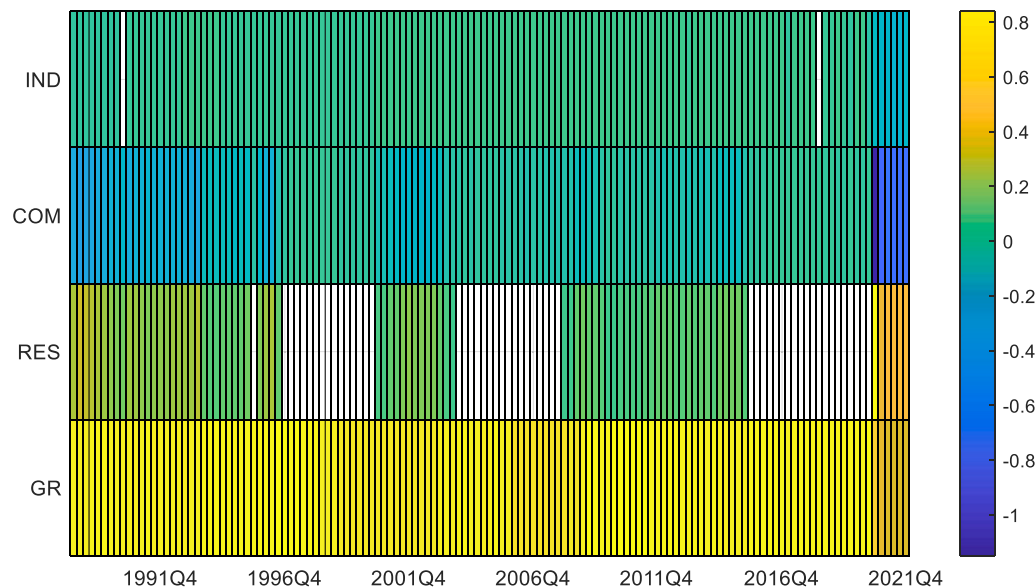


Fig. A1. Variable selection of different energy consuming sector.

## Appendix B. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.eneco.2023.107268>.

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